INCH-POUND

MIL-S-82850(OS)
5 February 1990
SUPERSEDING
WS 23739A
14 December 1987
(See 6.6)

MILITARY SPECIFICATION

SPRING, BELLEVILLE WASHER, ARM/ENABLE SWITCH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, test, manufacture, and acceptance requirements for the Belleville Spring Washer, referred to herein as the spring.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-46850 Steel: Bar, Plate, Sheet, Strip, Forgings, and Extrusions, 18 Percent Nickel Alloy, Maraging 200 KSI, 250 KSI, and 300 KSI High Quality

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1338

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STANDARDS

FEDERAL

FED-STD-151 Metal, Test Methods

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

6204107 Washer, Spring Tension, Arm/Enable Switch

(Application for copies of NAVSEA drawings should be addressed to the Commanding Officer, Naval Ordnance Station, Attn: Code 3720, Indian Head, MD 20640-5000.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Commercial Packaging (DoD adopted)

ASTM E 8 Tension Testing of Metallic Materials (DoD adopted)

ASTM E 18 Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials (DoD adopted)

ASTM E 45 Determining the Inclusion Content of Steel (DoD adopted)

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification 6000 Series

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Performance characteristics. The load requirements of the springs shall meet the following when 31 springs are stacked alternately concave up, concave down, on a shaft having a diameter of 0.1230 ± 0.0002 inch. The stack of springs on the shaft shall be held to a stack height of 0.705 ± 0.005 inch for 30 to 45 minutes and then shall have the following loads at the following heights:

Stack height (inches)	Loads (pound-force, lbf)
0.995 ± 0.001	15 minimum
0.895 ± 0.001	60 minimum
0.760 ± 0.001	75 to 145

3.3 Design and construction. The spring shall meet the design dimensions of drawing 6204107 and material requirements below.

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3.3.1 Material. The spring shall be manufactured using material conforming to MIL-S-46850, Type III, grade 300, except that the chemical analysis, heat treatment, mechanical properties, hardness, grain size, and inclusion content shall be as specified below.

3.3.1.1 Chemical analysis. The steel chemical analysis shall be as specified in MIL-S-46850 for Type III, Grade 300, except that the cobalt content shall be 11.0 to 12.5 weight percent and the titanium content shall be 1.35 to 1.45 weight percent.

3.3.1.2 Heat treatment. Before scragging (see 6.4.1), the springs shall be heat treated at 925 ± 15°F in an air-circulating furnace and held at temperature for 3 hours ± 15 minutes. The springs shall then be air cooled to room temperature (77 ± 20°F), cleaned with trichloroethane or equivalent, and scragged. Three sheet tensile test specimens or three coil-strip tensile test specimens, made from the same heat of material as the springs, in accordance with ASTM E 8, shall accompany each heat treated lot. After scragging, the springs shall be cleaned with trichloroethane or equivalent, and steam oxide treated (see 6.4.2) at 925 ± 15°F for 3 hours ± 15 minutes and then cooled to room temperature.

3.3.1.2.1 Mechanical properties. The tensile yield strength at 0.2 percent offset shall be 330,000 pounds per square inch (psi) minimum and the elongation in 2 inches shall be 2 percent minimum.

3.3.1.3 Hardness. The hardness shall be Rockwell C-56 to C-59.

3.3.1.4 Grain size. The grain size shall be 7 or finer.

3.3.1.5 Inclusion content. There shall be no A, B, C, or D inclusions permitted. Titanium carbonitrides, referred to as E Series, shall have 1.5 maximum thin series and 1.0 maximum thick series, as defined in ASTM E 45.

3.3.2 Scragging. The springs shall be scragged twice, after the heat treat and before steam oxide heat. The springs shall meet the spring height specified on drawing 6204107 after the final scragging operation.

3.3.3 Workmanship. The springs shall be free from surface contamination, tool marks and other imperfections which may adversely affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The contractor shall test samples to ensure that the requirements of this specification are met. A sample of 200 springs shall be subjected to first article inspection.

4.4 Quality conformance inspection. Quality conformance testing shall consist of the tests specified below in 4.5 to ensure the springs meet the requirements specified herein.

4.4.1 Lot. A lot shall be material of the same heat, form, and size, which has been processed through the same manufacturing operations at the same time.

4.5 Methods of inspection. The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

4.5.1 Dimensional inspection. The spring samples shall be inspected for dimensional conformance to drawing 6204107 as required in 3.3 and 3.3.3.

4.5.2 Lot inspection and acceptance tests. Two sets of 31 springs shall be stacked and tested to verify compliance with 3.2. Dimensional requirements shall be verified in accordance with MIL-STD-105, single sampling, normal inspection, Level II, with an acceptable quality level (AQL) of 2.5 percent.

4.5.3 Material inspection. The spring samples shall be inspected for conformance to the material requirements of MIL-S-46850 as specified in 3.3.1.

4.5.3.1 Chemical analysis. Chemical analysis shall be performed on samples in accordance with FED-STD-151, Method 111 or 112 and meet the requirements of 3.3.1.1.

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4.5.3.2. Mechanical properties. Testing of mechanical properties shall be in accordance with ASTM E 8 and meet the requirements of 3.3.2.1.

4.5.3.3 Hardness tests. Hardness tests shall be performed in accordance with ASTM E 18 to meet the requirements of 3.3.1.4.

4.5.3.4 Grain size. Specimen from each lot of steel tested in accordance with metallographic methods to meet the requirements of 3.3.1.5.

4.5.3.5 Inclusion content. Inclusion content shall be determined in accordance with ASTM E 45 and shall meet the requirements of 3.3.1.6.

4.5.3.6 Scragging. Scragging (see 6.4.1) shall be performed on sample springs to meet the requirements of 3.3.2.

4.6 Inspection of packaging. The spring containers and container marking shall be visually examined for compliance with section 5.

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C or in accordance with standard commercial practice as specified herein. The unit pack quantity shall be one lot of springs or 10 pounds, whichever is less.

5.1.1 Level C. The springs shall be packaged to afford adequate protection against loss, deterioration and damage during shipment from the supply source to the first receiving activity. Containers in the same shipment shall be of the same size. The packaging shall conform to National Motor Freight Classification, and UFC 6000 or to other carrier rules and regulations as applicable to the mode of transportation.

5.1.2 Commercial packaging. The springs shall be packed in accordance with standard commercial practice and the applicable requirements of ASTM D 3951.

5.2 Marking. Spring containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The spring is intended for use in the Standard Missile Mark 104 Dual Thrust Rocket Motor Arm/Enable Switch to provide the necessary loads on the arming lever for the life of the switch.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Packaging level or type as specified in 5.1.1.

d. Marking, if other than as specified in 5.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Scragging. Scragging is a process of compressing the springs to the flat position after the springs have been coned and heat treated (prior to steam oxide treatment).

6.4.2 Steam oxide. Steam oxide or steam treating is a method of tempering and produces a useful oxide coating. This coating is resistant to scoring of moving parts, extends the wear life of the parts, and improves the resistance of the material to corrosion.

6.5 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.6 Supersedure information. MIL-S-82850 incorporates the following engineering change proposal (ECP) and notice of revision (NOR):

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ECP		NOR	
MTA083	(11/30/88)	MTA083.2	(11/30/88)

Preparing Activity:
NAVY-OS
(Project 1338-N015)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-82850	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
SPRING, BELLEVILLE WASHER, ARM/ENABLE SWITCH

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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